SASHA BOGURAEV

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Last Updated: November 9, 2026

Education

University of Texas at Austin, Austin, USA
Ph.D. in Computational Linguistics; 2024 – Present (Expected: May, 2029)
Focus: Computational Linguistics and Language Model2 Interpretability.
Advisor: Dr. Kyle Mahowald

Cornell University, Ithaca, NY, USA
B.A. *Summa Cum Laude*; Computer Science, Mathematics, College Scholar; 2020-2024.
Thesis: Using Multi-Agent Reinforcement Learning to Study Language Evolution.
Committee: Dr. Morten Christiansen (Chair), Dr. Eugene Vinitsky (NYU), Dr. Veit Elser

Peer-reviewed Publications

*: Equal contribution

2026 *Denis Sutter, *Qing Yao, ***Sasha Boguraev**, Julian Minder, Antonio di Maio, Thomas Hofmann, Kyle Mahowald, Tiago Pimentel. Parasite Features: Causal Abstraction Through Spurious Pathways. *In Advances in Neural Information Processing Systems 39 (NeurIPS 2026)*. Paris, France. **Accepted.**

2026 Qing Yao, **Sasha Boguraev**, Tiago Pimentel, Kyle Mahowald. The Emergence and Sparsification of a Syntactic Circuit. *In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*. Budapest, Hungary. **Accepted; Best Paper Nominee.**

2026 **Sasha Boguraev**, Kyle Mahowald. Causal Drawbridges: Characterizing Gradient Blocking of Syntactic Islands in Transformer LMs. *In the Third Conference on Language Modeling (COLM 2026)*. San Francisco, USA.

2026 ***Sasha Boguraev**, *Qing Yao, Kyle Mahowald. France or Spain or Germany or France: A Neural Account of Non-Redundant Redundant Disjunctions. *In Proceedings of the 48th Annual Meeting of the Cognitive Science Society (CogSci 2026)*. Rio de Janeiro, Brazil.

2025 **Sasha Boguraev**, Christopher Potts, Kyle Mahowald. Causal Interventions Reveal Shared Structure Across English Filler–Gap Constructions. *In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*. Suzhou, China. **Outstanding Paper; Oral Spotlight.**

2025 **Sasha Boguraev**, Katrin E. Erk, Kyle Mahowald, James Shearer, Steve M. Wechsler. Reinforcement Learning Produces Efficient Case-Marking Systems. *In Proceedings of the 47th Annual Meeting of the Cognitive Science Society (CogSci 2025)*. San Francisco, USA.

2024 **Sasha Boguraev**, Ben Lipkin, Leonie Weissweiler, Kyle Mahowald. Models Can and Should Embrace the Communicative Nature of Human-Generated Math. *In the 4th Workshop on Mathematical Reasoning and AI (MATH-AI) at NeurIPS 2024*. Vancouver, Canada.

Work In Progress and Preprints

2026 **Sasha Boguraev**, Toshiki Nakai, Kyle Mahowald, Julius Steuer. Causal Interventions Reveal Typologically Organized Syntactic Mechanisms in Multilingual Language Models. *Under Review*.

2026 Elias Stengel-Eskin, Newton Sander, Carlos Bonetti, **Sasha Boguraev**, James Bowler, Hale Sirin, Simon Kirby. GlossoGen: Emergent Language in Complex Multi-Agent LLM Interactions. *Under Review*.

2026 **Sasha Boguraev**, Qing Yao, Kanishka Misra, Kyle Mahowald. Language Models and Experimental Syntax. In G. Goodall (Ed.), *The Cambridge Handbook of Experimental Syntax* (2nd ed., Ch. 26). Cambridge University Press. *Invited chapter, submitted*.

Peer-reviewed Abstracts

2027 **Sasha Boguraev**, Toshiki Nakai, Kyle Mahowald, Julius Steuer. Causal Interventions Provide Evidence of Cross-Lingual Representation Convergence in Language Models. *Annual Meeting of the Linguistic Society of America*. San Francisco, USA. **Oral; Accepted**.

2026 **Sasha Boguraev**, Kyle Mahowald. Causal Drawbridges: How Transformers Strand and Unstrand Syntactic Islands. *25th Annual Meeting of the Texas Linguistics Society*. Austin, USA. **Oral**.

Awards

2026 **Best Paper Nominee**. *EMNLP 2026*.

2025 **Outstanding Paper**. 8 Outstanding Paper Awards out of 8,174 Submitted Papers (Top 0.1%.) *EMNLP 2025*.

2024 **Professional Development Award**. UT Austin College of Arts and Sciences. Amount: $1,200.

2024 **Cornell Arts & Sciences Extraordinary Senior**.

2023 **Cornell Computer Science Course Staff Exceptional Service Award**.

2023 **California Institute of Technology, Summer Undergraduate Research Fellowship**. Amount: $10,000.

2021 **Cornell Robert S. Harrison College Scholar**

Invited Talks

Fall 2026 *Understanding the Internal Structure of Linguistic Abstractions in Neural Language Models.* Georgetown PICoL Lab. Washington, D.C., USA.

Fall 2026 *Understanding the Internal Structure of Linguistic Abstractions in Neural Language Models.* MIT Computational Psycholinguistics Lab. Cambridge, USA.

Conference and Workshop Talks

Spring 2026	*Causal Drawbridges: How Transformers Strand and Unstrand Syntactic Islands.* UT Austin Harrington Fellows Symposium 2026. Austin, USA.
Fall 2025	*Causal Interventions Reveal Shared Structure Across English Filler–Gap Constructions.* EMNLP 2025. Suzhou, China. **Oral Spotlight.**
Fall 2025	*Causal Interventions Reveal Shared Structure Across English Filler–Gap Constructions.* First Workshop on the Interplay of Model Behavior and Model Internals: A Workshop at the Conference on Language Modeling (COLM) 2025. Montreal, Canada. **Oral Spotlight.**
Fall 2025	*Causal Interventions Reveal Shared Structure Across English Filler–Gap Constructions.* Workshop on LLMs and the Brain, Rice University. Houston, USA.

Service

- **Co-Organizer:** BlackboxNLP 2027 – The Tenth Workshop on Analyzing and Interpreting Neural Networks for NLP, co-located with EMNLP 2027.
- **Co-Organizer:** South-by-Semantics Workshop (SxSW), University of Texas at Austin. 2025 - Present.
- **Co-Organizer:** Texas Linguistics Society, 2026.
- **Graduate Student Representative:** 2025 Search Committee for Assistant Professor in Computational Linguistics. Department of Linguistics, UT Austin.

Teaching

Guest Lecture - *Introduction to Language*, (LIN 306), UT Austin.
Semester: Fall 2025, Spring 2026.

Teaching Assistant - *Natural Language Processing* (CS 4740), Cornell University.
Semester: Fall 2023
Course Professor: Dr. Lillian Lee
Developed two (of four) course assignments along with a team of four other TAs. Held two hours of office hours per week, assisted in grading of all homeworks and exams. ($\approx$ 400 students enrolled).

Reviewing

Conference	EMNLP 2026; ACL 2026; EACL 2026, 2027.
Workshop	Interp4Discovery 2026; BlackboxNLP 2026; Workshop on the Interplay of Model Behavior and Internals, 2025.

Industry Experience

Summer 2023	**NASA Jet Propulsion Laboratory** - *Summer Undergraduate Research Fellow (Caltech SURF)* **Project:** Prototyping Information Retrieval and Retrieval-Augmented Generation (RAG) Systems for Space Flight Project Management.

Professional Affiliations

- Linguistic Society of America (LSA)
- Association for Computational Linguistics (ACL)
- Cognitive Science Society (CogSci)
- Mathematical Association of America (MAA)